

**Savitribai Phule Pune University**  
**T.Y.B.Sc.(C.S.) (NEP – 2020) SEM – V Practical Examination**  
**Lab Course CS-305-MJ-P**  
**Operating System-I**

**Duration: 3 Hours**

**Maximum Marks:35**

-----

Q.1 Write the simulation program for demand paging and show the page scheduling and total number of page faults according the optimal page replacement algorithm. Assume the memory of 'n' frames.

Reference String: 8, 5, 7, 8, 5, 7, 2, 3, 7, 3, 5, 9, 4, 6, 2

[15 Marks ]

Q.2 Write the program to simulate Non-preemptive Shortest Job First (SJF) scheduling. the arrival time and first CPU burst for different n number of processes should be input to the algorithm. The next CPU burst should be generated randomly. The output should give Gantt chart, turnaround time and waiting time for each process. Also find the average waiting time and turnaround time.

[15 marks]

Q.3. Oral/Viva

[5 Marks]